

Ahmadreza (Aaron) Nia

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SUMMARY

Machine Learning Engineer at Milwaukee Tool and final-year Ph.D. candidate with over six years of experience building signal processing and embedded AI pipelines. Strong background in time-series signal processing, sensor data pipelines, IoT connectivity, and deploying models under real compute and memory constraints on embedded devices, with a track record of working across teams from test engineers to development teams.

Areas of Focus: Signal Processing | Embedded Systems | IoT | Anomaly Detection | Edge Computing | Deep Learning

EDUCATION

University of Illinois at Urbana-Champaign (UIUC)

Ph.D. with Data Science concentration in Mechanical Engineering | GPA: 3.95

06/2021 – 11/2026

Advisors: Prof. Nahrstedt (Computer Science), Prof. Shao (Mechanics)

University of Illinois at Urbana-Champaign (UIUC)

M.Sc. in Mechanical Engineering | GPA: 3.93

06/2021 – 12/2023

Sharif University of Technology

B.S. in Mechanical Engineering | GPA: 3.72

08/2015 – 08/2020

SKILLS

Programming: Python (PyTorch, TFLite, Scikit-learn, TensorFlow, Keras, OpenCV, Pika, Seaborn, NumPy, Pandas), R, MATLAB, SQL, C++

General Software: MQTT, RabbitMQ, Raspberry Pi, Arduino, Node-RED, Azure, Git, Grafana, Linux, Spark, Docker

General Skills: Signal Processing, On-Device Inference, Time-Series Analysis, IoT, Edge-Cloud Computing, End-to-End ML Pipelines, Project Leadership, Technical Mentorship, Cross-Functional Collaboration, High-Performance Computing, Algorithm Design, Linear Algebra, RL, Quantization, Data Collection/Annotations, Embedded Systems, Problem Framing

Relevant Courses: IoT & Cognitive Computing, DL for Computer Vision, Machine Learning, ML in Physics, Optimization, Pattern Recognition, Statistical Learning, Data Science, Basics of Statistical Learning, Big Data

EXPERIENCE

Machine Learning Engineer

Milwaukee Electric Tool Corporation, Brookfield, WI

05/2025–08/2025 (Intern), 05/2026–Present (Full-time)

- Saved \$1M+ annually in BOM costs by architecting and deploying an end-to-end ML pipeline for unsafe-event detection under tight compute and memory constraints, achieving reliable performance on embedded devices.
- Achieved robust unsafe-event detection by engineering time-series signal preprocessing and feature-selection workflows in Python and benchmarking multiple ML techniques including XGBoost on Azure, resulting in 100% unsafe-event recall with a 4% false-alarm rate.
- Reduced inference computation by ~90% by optimizing model features and applying quantization for embedded C++ deployment, while meeting strict latency and memory constraints with negligible performance loss.
- Demonstrated production readiness in real-world conditions by deploying models on MCU and delivering live, on-tool demonstrations for cross-functional teams.
- Evaluating wireless connectivity and data throughput requirements for connected jobsite devices, turning findings into infrastructure and data-collection recommendations for cross-functional stakeholders.

Research Assistant

Multimedia Operating Systems and Networking, UIUC, IL ([Paper](#))

06/2022 – Present

- Enabled pre-print G-code anomaly detection by building an LLM-agent workflow for context-aware defect analysis, with potential to reduce material waste and print time.
- Enabled systematic and quantitative evaluation of LLMs for additive manufacturing by developing a specialized benchmark dataset and expert-validated evaluation protocols.
- Produced expert-driven benchmarking results by evaluating leading closed- and open-source LLMs on defect identification, technical reasoning, and task execution, with reviews from 14 domain experts.

Coordinated Science Laboratory (CSL), UIUC, IL (Paper) (Code)	06/2022 – Present
- Achieved an 8.08% accuracy improvement in domain generalization by developing a distributed learning framework with a novel loss function, outperforming state-of-the-art methods while preserving data privacy. - Demonstrated 66% data efficiency and 50% time efficiency over baseline methods through evaluation of the proposed framework in a realistic edge–cloud emulation using Raspberry Pi and RabbitMQ.	
Automation and Digital Manufacturing Lab, UIUC, IL (Paper) (News)	05/2022 – 08/2023
- Achieved real-time fault detection on edge devices by implementing an end-to-end deep learning pipeline for welding machines, processing high-dimensional time-series data in 385 ms per welding cycle. - Enhanced tool condition classification accuracy by 8.3% using a data augmentation approach to address dataset shift.	
Health Care Engineering Systems Center (HCESC), UIUC, IL (Paper)	06/2021 – 11/2022
- Showcased potential for quantitative evaluation of stroke rehabilitation by classifying activities of daily living. - Achieved 99% classification accuracy by developing data-driven SVM and CNN models using Leap Motion sensor data. - Improved classification accuracy by 11% using spatiotemporal feature extraction techniques.	

ACHIEVEMENTS AND AWARDS

Awarded an NSF Travel Grant for the 53rd North American Manufacturing Research Conference, United States	07/2025
Received the Best Paper Award at UEMCON IEEE 2023 conference, United States	08/2023
Awarded UIUC Doctoral Fellowship, United States	05/2021
Led a team to first place in the national "Tanabar" Competition, Tehran	02/2019
Ranked 60 th among 180,000 participants in the national university entrance exam (Konkur), Tehran	07/2015

SERVICES

Served as a reviewer for 26 AI/ML-track papers in 6 journals and 4 conferences	06/2023 – Present
Instructed a graduate-level course and leading lab sessions for 60 students at UIUC	08/2023 – Present
Served as a teaching assistant for 6 different courses	08/2018 – Present
Mentored a student in NCSA's summer Research Experiences for Undergraduates (REU) program	05/2023 – 08/2023

SELECTED PUBLICATIONS

- Eslaminia, A., Cai, C., Smith, C., Mei, R. S., Li, S., Malhotra, R., ... & Shao, C. (2026). *LLM-ADAM: A Generalizable LLM Agent Framework for Pre-Print Anomaly Detection in Additive Manufacturing*. arXiv preprint arXiv:2605.03328.
- Eslaminia, A., Jackson, A., Tian, B., Stern, A., Gordon, H., Malhotra, R., ... & Shao, C. (2025). *FDM-bench: a domain-specific benchmark for evaluating large language models in additive manufacturing*. Manufacturing Letters, 44, 1415-1424.
- Eslaminia, A., Meng, Y., Nahrstedt, K., & Shao, C. (2024). *Federated domain generalization for condition monitoring in ultrasonic metal welding*. Journal of Manufacturing Systems, 77, 1-12.
- Eslaminia, A., Lu, K. C., Nahrstedt, K., & Shao, C. (2026). *Adaptive Unknown Fault Detection and Few-Shot Continual Learning for Condition Monitoring in Ultrasonic Metal Welding*. arXiv preprint arXiv:2604.13465.
- Tian, B., Eslaminia, A., ..., & Nahrstedt, K. (2023, October). *Weldmon: A cost-effective ultrasonic welding machine condition monitoring system*. In 2023 IEEE 14th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON) (pp. 0310-0319). IEEE.
- Yu, Y., Wu, M., Eslaminia, A., Zhao, L., Yan, K., & Nahrstedt, K. (2026). *QoS-QoE Translation with Large Language Model*. arXiv preprint arXiv:2604.08703.
- Li, S., Eslaminia, A., & Shao, C. (2025). *Multi-Modal Zero-Shot Prediction of Color Trajectories in Food Drying*. arXiv preprint arXiv:2512.06190.
- Tian, B., Wang, Y., Eslaminia, A., Sardela, M., Dallesasse, J., & Nahrstedt, K. (2025). *MachineStethoscope: A smart and cost-effective machine health monitoring system*. In 2025 IEEE 8th International Conference on Multimedia Information Processing and Retrieval (MIPR) (pp. 323–329). IEEE
- Sharif, H., Eslaminia, A., Chembrammal, P., & Kesavadas, T. (2022). *Classification of activities of daily living based on grasp dynamics obtained from a leap motion controller*. Sensors, 22(21), 8273.
- Nasrabadi, A. M., Eslaminia, A. R., ... & L., & Behzadipour, S. (2022). *A new scheme for the development of IMU-based activity recognition systems for telerehabilitation*. Medical engineering & physics, 108, 103876